

Auteur: Rob Willemsen
 Studierichting: Informatica
 Oefeningenreeks 4A nr. 41.13

$$\begin{aligned}
 & \int \frac{x-1}{\sqrt{x^2+5x+6}} dx \\
 &= \int \frac{\frac{1}{2}(2x+5) - \frac{7}{2}}{\sqrt{x^2+5x+6}} dx \\
 &= \frac{1}{2} \int \frac{2x+5}{\sqrt{x^2+5x+6}} dx - \frac{7}{2} \int \frac{1}{\sqrt{x^2+5x+6}} dx \\
 & t = x^2 + 5x + 6 \Rightarrow dt = (2x+5)dx \\
 &= \frac{1}{2} \int \frac{dt}{\sqrt{t}} - \frac{7}{2} \int \frac{dx}{\left(x + \frac{5}{2}\right)^2 - \frac{1}{4}} \\
 &= \frac{1}{2} \int t^{-\frac{1}{2}} dt - \frac{7}{2} \int \frac{dx}{\left(x + \frac{5}{2}\right)^2 - \frac{1}{4}} \\
 &= \frac{1}{2} \cdot \frac{2}{1} t^{\frac{1}{2}} - \frac{7}{2} \ln \left| \left(x + \frac{5}{2}\right) + \sqrt{\left(x + \frac{5}{2}\right)^2 - \frac{1}{4}} \right| + C \\
 &= \sqrt{x^2+5x+6} - \frac{7}{2} \ln \left| x + \frac{5}{2} + \sqrt{x^2+5x+6} \right| + C
 \end{aligned}$$

References